

Thermal Properties of Matter

>> **Temperature** - Relative measure or indication of hotness or coldness.

It is the thermal state of a body which decides the direction of flow of heat energy by virtue of temperature difference.

>> SI unit = Kelvin [K].

>> **Heat** - Energy in transit that flows from one body to another due to temp. difference.

S.I unit = Joule.

C.G.S unit = Calorie.

Heat	Temperature
Total K.E	Avg. K.E
It flows from high to low temperature	It decides the direction of flow.

>> Heat - Non mechanical way of transfer of Heat.
work - mechanical

>> $\text{Donework} \propto \text{Heat produced}$

$$w \propto Q$$

$$w \propto JQ$$

→ Joule's mechanical equivalence of heat.

>> measurement of Temperature → Thermometry.

- Temperature is (measurement) measured by thermometer.
- Properties used as basis of constructing thermometers:-

Length of liquid column, pressure & volume of liquid.

- For constructing any standard scale, two fixed references need

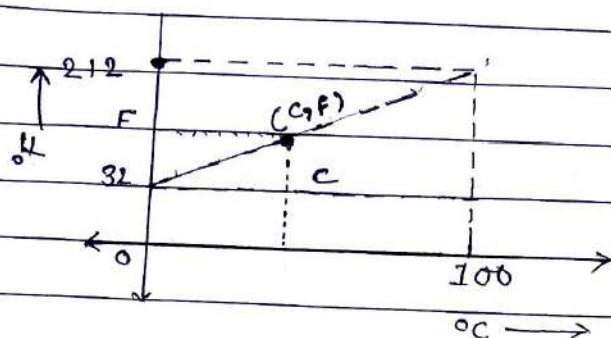
<1> Boiling of water. (Steam point)

<2> melting of ice. (Ice point)

$$^{\circ}\text{C} = 0^{\circ} \rightarrow 100^{\circ}\text{C}$$

$$^{\circ}\text{F} = 32^{\circ} \rightarrow 212^{\circ}\text{F}$$

$$\text{K} = 273.15 \rightarrow 373.15 \text{ K}$$



Similarity of triangles

$$\frac{F-32}{212-32} = \frac{C-0}{100-0} \quad \text{constant}$$

$$F-32 = \frac{180}{100} C$$

$$\Rightarrow C = \frac{5}{9} (F-32)$$

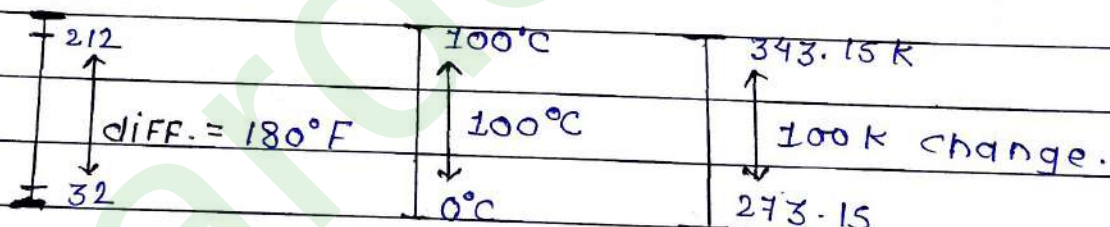
>> For conversion of any scale

$$K = 273.15 + ^\circ C$$

$$\frac{\text{Reading} - \text{icepoint / Lowpoint}}{\text{Boiling Point} - \text{Ice point}} = \text{constant}$$

Absolute zero

$$0K = -273^\circ C = -460^\circ F$$



$$1^\circ C \text{ change} = 1 \text{ kelvin change} = 1.8 F \text{ change}$$

>> Boyle's Law: $PV = \text{constant}$ (At constant T)

>> Charles's Law: $\frac{V}{T} = \text{constant}$ ($P = \text{const}$)

$$\Delta C = \Delta K = \frac{5}{9} \Delta F$$

>> Ideal Gas Equation:-

$$PV = nRT$$

→ universal gas constant ($R = 8.31 \text{ J/mol}\cdot\text{K}$)

Steam point	373.15 K	100°C	212°F
Ice point	273.15	0°C	32°F
Absolute zero	0 K	-273.15°C	-459.69°F

↳ No motion in molecules

>> Temperature, at which the reading in centigrade and Fahrenheit scale is same

$$\begin{array}{ccc} ^\circ\text{C} & ^\circ\text{F} & \Rightarrow T - 0 = T - 32 = \frac{T}{100} = \frac{T - 32}{180} \\ 0^\circ\text{C} & 32^\circ\text{F} & 100 - 0 \quad 212 - 32 \quad 100 \quad 180 \end{array}$$

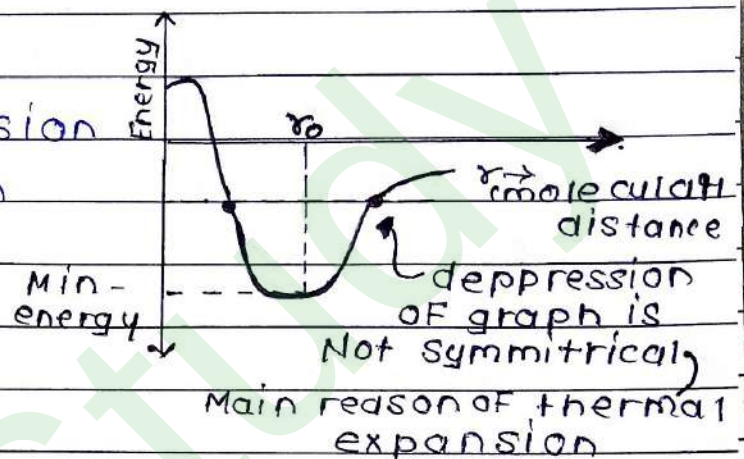
$$\begin{array}{ccc} 100^\circ\text{C} & 212^\circ\text{F} & \Rightarrow \frac{18T}{10} = T - 32 \Rightarrow T - \frac{9T}{5} = 32 \Rightarrow \frac{5T - 45T}{5} = 32 \\ T & T & \end{array}$$

$$\Rightarrow \frac{40T}{5} = 32 \Rightarrow T = 4^\circ\text{C}$$

Thermal Expansion

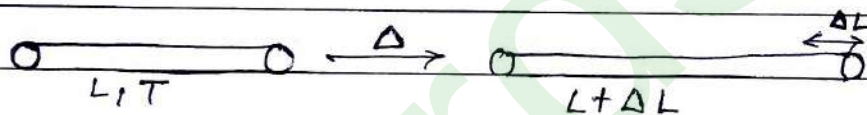
> The increase in the dimensions of body due to the increase in its temperature is called Thermal Expansion.

>> On heating $\rightarrow$ Expansion
On cooling $\rightarrow$ contraction



(i) Linear Expansion

(Expansion in the length)



$$\Delta L \propto \Delta T$$

$$\Delta L \propto L$$

$$\Delta L = L \alpha \Delta T$$

$$\alpha = \frac{\Delta L}{L \Delta T} \text{ (linear strain) per unit change in } T$$

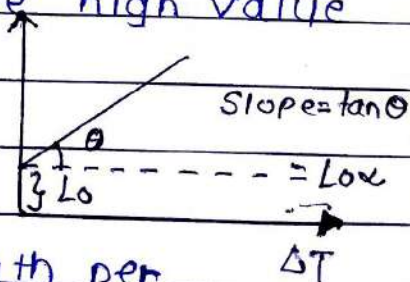
$\rightarrow$ coefficient of linear expansion or linear expansivity (depends upon material & temp.)

$\rightarrow$ High value of α = more expansion

$\rightarrow$ e.g. Metals expand more hence have high value of α

IF $L = 1\text{m}$, $\Delta T = 1\text{ unit}$, then

$$\alpha = \Delta L$$



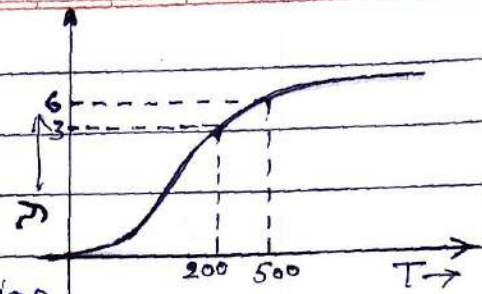
α = Increase in length per unit length per unit rise in temp.

$$\alpha_{\text{composite}} = \frac{l_1 \alpha_1 + l_2 \alpha_2}{l_1 + l_2}$$

iii) Volume Expansion / Superficial ($\Delta T < 100^\circ\text{C}$)

$$\Delta V = V\gamma\Delta T$$

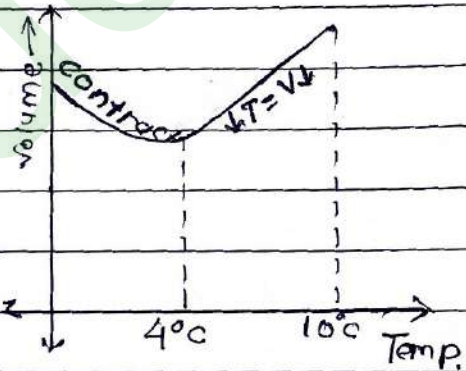
→ coefficient of vol. expansion



- In a non-homogenous object $\Rightarrow \gamma = \alpha_x + \alpha_y + \alpha_z$
- γ depends largely upon temperature.
At low temp, γ increase with temp but at high temperature γ becomes constant.

• Anomalous Expansion of water Reason: H-bonding.

- water contracts on heating between 0°C and 4°C
- water has max. density at 4°C .



Causes

- Due to this property of water, many phenomena are observed:-

- In cold countries water pipes burst when temperature decrease, because below 4°C water contracts.
- In cold nights, crops destroy.
Ice freezes only on the surface of lakes and rivers.

>> For an ideal gas, Find the value of γ at constant Pressure :-

$$PV = nRT$$

$$PdV = nRdT$$

$$PV\gamma dT = nRdT \Rightarrow \gamma = \frac{nR}{PV} = \frac{nR}{nRT} \Rightarrow \gamma = \frac{1}{T}$$

$$\text{Volume} = a^3 (\text{length})^3 \quad \left[\begin{array}{l} \Delta L = L \alpha \Delta T \\ dL = L \alpha dT \end{array} \right]$$

$$\int v = \int L^3$$

$$dv = 3L^2 dL$$

$$3L^2 dL = dv$$

$$3L^2 dL = L$$

$$(dv = v \gamma dT)$$

$$dv = v \gamma dT$$

$$3L^2 dL = L^3 \gamma dT$$

$$(v = L^3)$$

$$3dL = L \gamma dT$$

$$3(L \alpha dT) = L \gamma dT$$

$$3\alpha = \gamma$$

$$\text{e.g. } \alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$$

$$\gamma_{\text{steel}} = 3.6 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$$

$$\Rightarrow \boxed{\gamma = 3\alpha}$$

For amorphous solids

(Isotropic solids) $\rightarrow$ same in any direction.

(iii) Areal Expansion

(Anisotropic $\rightarrow$ crystalline)

$$\Delta A = A \beta \Delta T$$

β Coefficient of Areal Expansion

$$A = L^2$$

$$dA = 2L dL$$

$$A \beta dT = 2L \times L \alpha dT$$

$$L^2 \beta = 2L^2 \alpha$$

$\alpha, \beta, \gamma \rightarrow$ same dimension

$$\Rightarrow \boxed{\beta = 2\alpha}$$

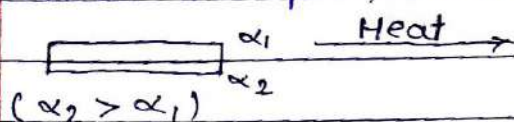
For isotropic solids

$\gg$ Triple point of H_2O - Point where H_2O exists in all the three states

i.e., 0.01°C or 273.16 K .

>> Bimetallic Strip

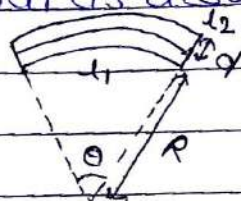
- When temp $\uparrow$, it bends towards less α



$$L_1 = L \alpha_1 \Delta T$$

$$L_2 = L \alpha_2 \Delta T$$

$$\Rightarrow \frac{L_2}{L_1} = \frac{\alpha_1}{\alpha_2} \quad \text{--- (I)}$$



$$L_1 = R\theta$$

$$L_2 = (R+d)\theta$$

$$\frac{L_2}{L_1} = \frac{R+d}{R} \quad \text{--- (II)}$$

compare (I) and (II)

$$\frac{\alpha_2}{\alpha_1} = \frac{1+d}{R} \Rightarrow \frac{d}{R} = \frac{\alpha_2 - \alpha_1}{\alpha_1} \Rightarrow$$

$$R = \frac{d \alpha_1}{\alpha_2 - \alpha_1}$$

>> Fault in metallic scale

$$\boxed{\text{Total Reading} = \text{measured Length} \pm \Delta L_{\text{scale}}}$$

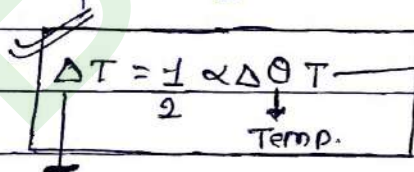
>> Pendulum Base clock

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\frac{\Delta T}{T} = \frac{1}{2} \left[\frac{\Delta L}{L} \right] \quad \left(\frac{\Delta L}{L} = \alpha \Delta \theta \right)$$

$$\frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta$$

Temp $\uparrow \rightarrow$ Length $\uparrow \Rightarrow$ Time $\uparrow$ = slow
(Time loss)



original Time (In one time period)

Loss/gain in time. $\Rightarrow$ In 't' time Loss/gain = $\frac{\Delta T}{T} \times t$

>> Effect on density of liquid

$$\rho' = \rho (1 - \gamma \Delta T)$$

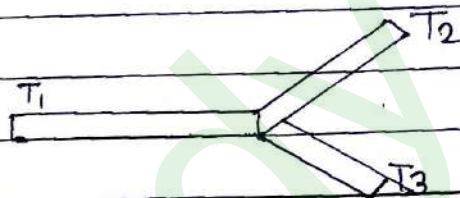
$$\text{or } \Delta \rho = \rho_0 \gamma \Delta T$$

>> Heat Gain = Heat loss

$$T_{\text{mix}} = \frac{m_1 s_1 T_1 + m_2 s_2 T_2}{m_1 s_1 + m_2 s_2}$$

>> Temperature at junction —

$$T_{\text{junction}} = \frac{T_1 A_1 k_1}{L_1} + \frac{T_2 A_2 k_2}{L_2}$$



$$\frac{A_1 k_1}{L_1} + \frac{A_2 k_2}{L_2}$$

Same area
Same length

$$T_{\text{junction}} = \frac{T_1 + T_2 + T_3}{3}$$

>> mixture of ice (m gram at 0°C) and water (w gram — water at $T^\circ\text{C}$)

$$mL = wT$$

$$T_{\text{mix}} = 0^\circ\text{C}$$

$$m_{\text{ice}} = 0$$

$$[m_{\text{water}} = mw]$$

$$ML > wT$$

$$T_{\text{mix}} = 0^\circ\text{C}$$

$$[m' = \frac{wT}{L}]$$

$$ML < wT$$

$$T = \frac{wT - mL}{w + m}$$

Thermal Stress

on heating, rod will try to expand

If compressive stress acts on rod

which compress rod by Δl

$$\Delta l_{\text{heat}} = \Delta l_{\text{stress compressive expansion}}$$

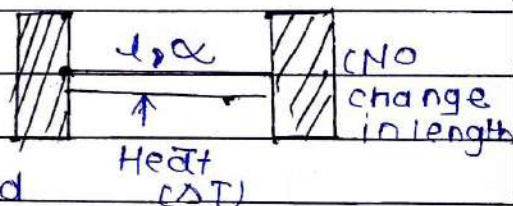
$$\text{Strain} = \frac{\Delta l}{l} = \alpha \Delta T = \alpha \Delta T$$

$$\text{Stress} = \tau \Rightarrow \text{Stress} = Y \text{ strain}$$

Strain

$$\frac{F}{A} = Y \alpha \Delta T \Rightarrow$$

$$F = YA \alpha \Delta T$$



Nature of different thermometers

1. Mercury thermometer (Thermal Expansion)

$[-38.8^{\circ}\text{C} - 356^{\circ}\text{C}]$ Range.

↓ Fill top by N_2 gas (high Pressure)

Higher point will increase to 500°C .

2. Gas thermometer (Thermal Expansion)

$[-260^{\circ}\text{C} - 16000^{\circ}\text{C}]$

- Pressure $\uparrow \rightarrow$ Temp. $\uparrow$

- High accurate but Very Slow.

3. Resistance thermometer

- Temp. $\uparrow$ — Resistance $\uparrow$ $\left[\frac{R_t - R_0}{t} = \frac{R_{100} - R_0}{100} \right]$

4. Thermometric thermometer

[Temp. change $\Rightarrow$ current change] Seebeck effect.

5. Pyrometer [Stephan's law]

Temp. of Sun, moon are measured

6. Baythoscope - measure temp. of ocean floor

7. Magnetic Thermometer

• measure the temp. below Kelvin.

Question Type-1

Given, $\alpha = 2 \times 10^{-3} \text{ K}^{-1}$

Δt (time) = T.

ΔT (temp) = 50 K

What will be the loss/gain in the time period of this pendulum?

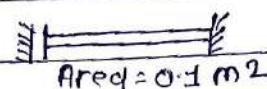
$$\frac{\Delta t}{t} = \frac{1}{2} \frac{\Delta L}{L} \quad \frac{1}{2} \alpha \Delta T = \frac{1}{2} \times 2 \times 10^{-3} \times 50 = \underline{\underline{0.05}}$$

Type II. Thermal physics + mechanic properties of Solid.

$$\gamma = 5 \times 10^4 \text{ Pa}$$

$$L_1 = 10 \text{ m}, \alpha = 4 \times 10^{-3} \text{ K}^{-1}$$

$$(\Delta \text{Temp.}) \text{ Rod} = 50 \text{ K}$$



① what is thermal strain $\left\{ \begin{array}{l} \text{actual} = 0 \\ \text{potential} = \frac{\Delta L}{L} = \alpha \Delta T = 4 \times 10^{-3} \times 50 \end{array} \right.$

② what is the thermal stress

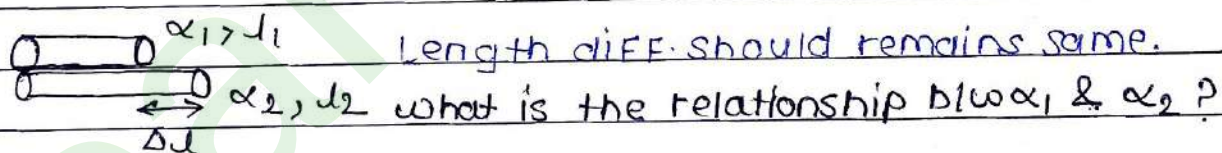
$$\begin{aligned} \text{Thermal stress} &= \gamma \cdot \text{Thermal strain} \\ &= 5 \times 10^4 \times (4 \times 10^{-3} \times 50) \text{ Pascals} \end{aligned}$$

③ Thermal tension developed in the rod.

$$P = \frac{F}{A} = \text{Thermal stress} = \frac{\text{Tension Developed}}{0.1}$$

$$\text{Tension} = 0.1 [5 \times 10^4 \times (4 \times 10^{-3} \times 50)]$$

Type III:- comparison Questions



$$\Delta L_2 - \Delta L_1 = l_{i2} - l_{i1}$$

$$[l_{i2} [1 + \alpha_2 \Delta T]] - [l_{i1} (1 + \alpha_1 \Delta T)] = l_{i2} - l_{i1}$$

$$\boxed{\alpha_1 l_{i1} = \alpha_2 l_{i2}}$$

[Equation in which difference in length does not depend on temperature]

* Thermal stress does not depend on length

$$\frac{F}{A} = \frac{\gamma \Delta L}{L} \left(\frac{\Delta L}{L} = \alpha \Delta T \right)$$

$$\boxed{\frac{F}{A} = \gamma \alpha \Delta T}$$

Specific Heat capacity $\Rightarrow$ Low $\Rightarrow$ Better conductor

(Not much Heat required)

$c \rightarrow \infty \Rightarrow T = \text{constant}$ (e.g. Heat Reservoir)

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Ajanta

Calorimetry

- Branch of physics that deals with heat and temp. change as well as changes in state of matter
- When we give heat to a substance two things will happen
 - (i) State of body will change
 - (ii) Temperature of body will change.

Thermal/heat capacity (C)	Specific Heat capacity (c_s) or (S)	Molar Heat capacity (C_m)
@ (Heat) added to a Body to raise temp. by $\pm 1K$	@ added to a unit mass to raise temp. by $\pm 1K$	@ added to one mole to raise temp. by $\pm 1K$
Depends on mass	Depends on material	
Unit: $J K^{-1}$	$J g^{-1} K^{-1}$ or $J kg^{-1} K^{-1}$	$J mol^{-1} K^{-1}$
$C = \frac{\Delta Q}{\Delta T}$	$C_s = \frac{Q}{m \Delta T}$	$C_m = \frac{Q}{n \Delta T}$
$\Rightarrow Q = C \Delta T$	$\Rightarrow Q = m c_s \Delta T$	$\Rightarrow Q = n C_m \Delta T$

$\Delta T = \text{Always positive}$

- In reality, c increases with temperature $\rightarrow Q = m \int_1^2 c dT$
- Calorimeter = Device in which heat measurement can be done
- $\rightarrow$ water has highest specific heat capacity ($4.2 J$ or 1 calorie)

Relations:-

$C \rightarrow$ Heat capacity

$$C = m c_s$$

c or $c_s \rightarrow$ Specific Heat capacity

$$C = n C_m$$

Principle of calorimetry / Law of mixtures = $\left[\begin{array}{l} \text{Heat Gained} = \text{Heat} \\ \text{by cool body} \quad \text{lost} \end{array} \right]$

By hot body.

Latent Heat

>> The amount of heat per unit mass transferred during change of state of the substance is called latent heat of the substance.

>> Latent heat is used to break the bonds, or weakened it.

Ice $\xrightarrow{\Delta}$ water [Latent heat of fusion]
(0°C) (0°C)

Water $\xrightarrow{\Delta}$ Steam [Latent heat of vapourisation]
(100°C) (100°C)

>> Specific latent heat $\Rightarrow$ latent heat per unit mass.

$$L = \frac{Q}{M}$$

$$\Rightarrow \boxed{Q = mL}$$

Latent heat

Specific latent heat.

Values to remember (H_2O)

(Cs)ice = 0.5 cal g⁻¹ K⁻¹

(Cs)water = 1 or 4.2 J

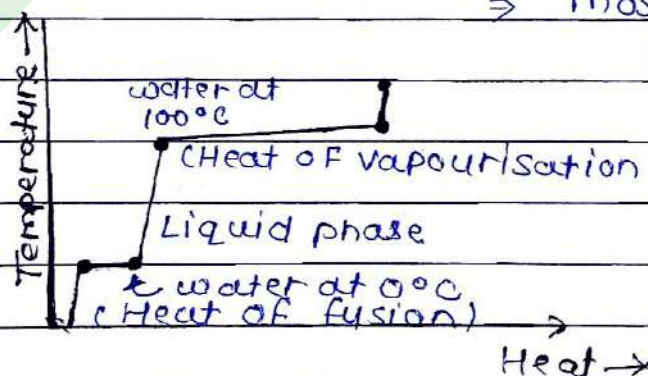
(Cs)steam = 0.47

$L_{\text{fusion}} = 80 \text{ cal g}^{-1}$

$L_{\text{vapourisation}} = 540 \text{ cal g}^{-1}$

$\Rightarrow$ mostly, $L_v > L_f$

(Steam's burns are more severe than H_2O burns)



Mixing Problems

Example-I $\Rightarrow$ On mixing 40g ice at 0°C with 20g water at 30°C . What will be the result.

- We have to take a reference point.

40g (ice) (0°C) + 20g (H_2O) (30°C)

$$Q = mL$$

$$= 40 \times 80$$

$$= 3200 \text{ H}_2\text{O} (0^\circ\text{C})$$

calorie

$$Q = mc\Delta T$$

$$= 20 \times 1 \times 30$$

$$= 600 \text{ calorie.}$$

20g (H_2O) (0°C) melts Ice.

Remaining heat = $3200 - 600 = 2600$ calorie.

All ice will not melt.

Let x ice will melt.

$$Q = mL$$

$$2600 = x \times 80 \Rightarrow x = \frac{60}{8} \text{ g}$$

$\text{H}_2\text{O} \rightarrow m_{\text{melt}} + 20\text{g.}$

Ice = $40 - m_{\text{melt}}$

Example-II

1g ice (-20°C) + 20g H_2O (40°C) (Find temp & composition = ?)

$$Q_1 = 1 \times 0.5 \times 20$$

$$= 10 \text{ calorie}$$

$$Q_2 = 1 \times 80$$

$$= 80 \text{ calorie.}$$

$$Q_1 + Q_2 = 90 \text{ calories}$$

$$Q_3 = 20 \times 1 \times 40 =$$

$$800 \text{ calorie.}$$

$$Q_{\text{extra}} = 710 \text{ calories}$$

composition = completely H_2O (21g)

Temperature of mixture $\Rightarrow Q = mc\Delta T$

$$710 = (21) (1) (\Delta T)$$

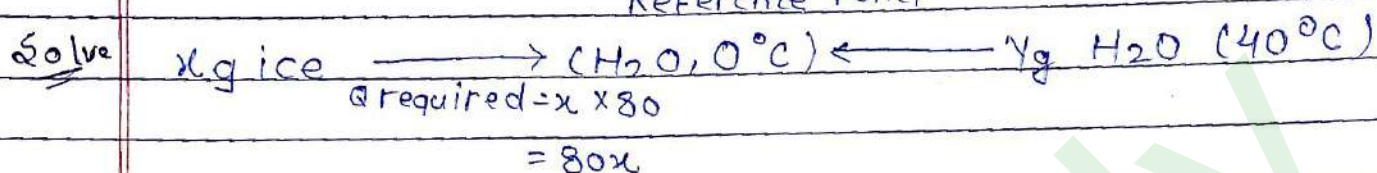
$$\Delta T = \frac{710}{21}^\circ\text{C}$$

$$\underline{\underline{31}}$$

Example - III :- x g Ice + y (g) H_2O ($40^\circ C$)

what should be $\frac{x}{y}$ if the entire composition is of H_2O at $0^\circ C$?

Reference Point



$$Q_{\text{reqd}} = Q_{\text{Released}}$$

$$80x = 40y \rightarrow \frac{x}{y} = \frac{1}{2}$$

Example - IV 20g ice ($0^\circ C$) + 1g steam ($100^\circ C$)

Temp. of mixture, composition of mixture = ?

$$Q_1 = 1 \times 540 = 540 \text{ calories}$$

$$Q_2 = 1 \times 1 \times 100 = 100 \text{ calorie}$$

$$Q_{\text{Required}} = 20 \times 80$$

$$= 1600 \text{ calorie}$$

H_2O ($0^\circ C$)

$$Q_{\text{Release}} = 640 \text{ calories}$$

$$640 = m_{\text{melt}} + 80$$

$$m_{\text{melt}} = 8g$$

1g H_2O

9g H_2O

$$21 - 9 = 12g \text{ ice}$$

Short Trick

mass of ice : mass of steam = 8:1 $\Rightarrow T_{\text{mix}} = 0^\circ C$

mass of ice : mass of steam = 3:1 $\Rightarrow T_{\text{mix}} = 100^\circ C$

HEAT TRANSFER

→ It is due to difference in temperature

→ Can be done in three ways -

(1) Conduction

(2) Convection

(3) Radiation

Heat moves but
not particle

e.g. Rod

Particles also
move

Natural

Forced

(Gravity Affects
convection)

→ Need a medium.

Heat is released
is form of radiat-
ion (EMW)

→ No Need of medium

→ Fastest mode.

► Conduction

Heat = Q 

Heat current = $\frac{dQ}{dt} = H = \frac{\Delta T}{R} \Rightarrow H = \frac{T_A - T_B}{R_{\text{rod}}}$
(Heat per unit time)

$\Rightarrow R_{\text{rod}} = \frac{L_{\text{rod}}}{k \cdot A_{\text{rod}}}$
(Thermal Resistance)

$H = \frac{\Delta T \cdot k \cdot A}{L}$
Area
Length

[k = thermal conductivity]

⇒ In series, $H_1 = H_2$

$R_{\text{com}} = R_1 + R_2$

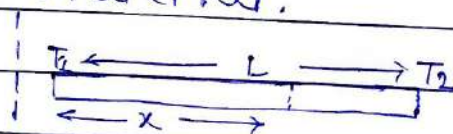
Dimension : $[MLT^{-3}K]$

$\left[\frac{L_{\text{comb}}}{k_{\text{comb}} \cdot A_{\text{comb}}} = \frac{L_1}{k_1 A_1} + \frac{L_2}{k_2 A_2} \right]$

→ k depends on nature of material.

⇒ In parallel,

$\frac{1}{R_{\text{comb}}} = \frac{1}{R_1} + \frac{1}{R_2}$



$\left[\frac{k_{\text{comb}} \cdot A_{\text{comb}}}{L_{\text{comb}}} = \frac{k_1 A_1}{L_1} + \frac{k_2 A_2}{L_2} \right]$

$T_x = T_1 - \left[\frac{T_1 - T_2}{L} \right] x$

➤ Radiation

- Heat emission/transfer due to high temperature of the body in the form of EMW.

- Bodies $\left\{ \begin{array}{l} \text{Black Bodies} \rightarrow \text{Power of Radiation } (P_1) \\ \text{Non Black Bodies} \rightarrow P_2 \end{array} \right. \rightarrow \boxed{P_1 > P_2}$ always

- Black body :-

IF a light is falling on the body things can happen :-

(1) absorption (a) (2) transmission (t) (3) Reflection (r)

$$\frac{Q_a \text{ heat absorbed}}{Q \text{ Heat falling}}, \quad \frac{Q_t}{Q}, \quad \frac{Q_r}{Q}$$

$$a + t + r = 1.$$

→ Black body is that whose absorbant = 1
(t, r = 0)

→ emissivity (e) of Non black body = $\frac{(P_{\text{Non black}})T}{(P_{\text{black body}})T}$

$e < 1 \rightarrow$ For non black body

$e = 1 \rightarrow$ For black body.

⇒ Stephan's Boltzmann's law

$$(\text{Intensity})_{\text{surface}} = \sigma T^4$$

↳ Stephan's Boltzmann's constant

$$\text{Power of black body} = \sigma T^4 \quad (\sigma = 5.67 \times 10^{-8} \text{ units})$$

Surface Area.

(unit = watt $\text{m}^{-2} \text{K}^{-4}$)

$$P_{BB} = (4\pi R^2) \sigma T^4$$

↓
(will remain same)

>> Solar Constant

(Intensity of Radiation from Sun on earth)

$$S_c = \frac{P_{\text{sun}}}{4\pi r^2} = \frac{(4\pi R^2)(\sigma T^4)}{4\pi r^2} = \sigma T^4 \left[\frac{R^2}{r^2} \right]$$

↓
[unit = $1.36 \times 10^3 \text{ watt s/m}^2$]

>> Wein's Displacement Law $\Rightarrow \lambda T = b$ (constant)

↓
(Weins constant = $2.8 \times 10^{-3} \text{ mK}$)

>> Weidman Franz Law: $\frac{K}{\sigma T} = \text{constant}$

>> Newton's Law of cooling $\Rightarrow$ Rate of heat loss $\propto$
Rate by which = $\left(-\frac{dT}{dt} \right) \propto \Delta T$
tem. drops

$t = \text{time}$
 $T = \text{temp}$

$$\left[\frac{(T_{\text{drop}})_t}{t} \propto (T_{\text{avg}} - T_{\text{surrounding}}) \right]$$

$$\left\{ \frac{T_1 - T_2}{t} = \propto \left[\frac{T_1 + T_2}{2} - T_0 \right] \right\}$$